

BUG REPORT · UNFILED DRAFT

Payment frequency premium goes stale under rapid selection

Switching the payment-frequency selector (month / fortnight / quarter / half year / year) several times in quick succession can leave the displayed premium showing a value that belongs to a previously-selected frequency, while the selector label itself has already moved on and shows the current one.

SEVERITY	COMPONENT	ENVIRONMENT	FOUND VIA
▲ High	Quote > Payment frequency	outsystems-dev (NBNZAAQuote)	Automated pressure test (Playwright)

REPORTED
2026-08-24 · Samuel Wilson

SUMMARY

The Quote step recalculates the premium via an async call (`ActionCallPricingEngineNZAA`) every time the payment frequency is changed. When a user changes frequency again before the previous recalculation has returned, the requests can resolve out of order. Whichever response lands *last* overwrites the premium on screen — even if it belongs to a frequency the user is no longer looking at. The frequency label is not affected (it updates instantly and correctly on click), so the result is a screen that confidently displays a price next to a frequency it doesn't match.

This was found while pressure-testing the quote flow for exactly this scenario, and reproduced reliably enough across repeated runs to be confident it's a real race, not a one-off fluke.

PRECONDITIONS

Start a new quote at </NBNZAAQuote/> , no login required. Complete the eligibility, personal details, AA membership and cover selection steps with any valid placeholder data (e.g. residency: Yes, DOB making the applicant an adult, AA member: No, cover: Life Cover) to reach the **Quote** step, then enter a cover amount of 500,000 . This settles on a baseline premium of \$45.83 /month .

STEPS TO REPRODUCE

With the Quote step showing \$45.83 /month (see **Evidence A** below), click the frequency selector (the black pill next to the premium) and choose **fortnight**. Wait for it to settle.

Confirms the correct value for comparison: \$21.15 /fortnight (**Evidence B**).

Switch back to **month** and let it settle again (\$45.83 /month), so the next step starts from a known state.

Now reproduce the race: open the frequency selector and click **year**, then **immediately** reopen it and click **month**, then **immediately** reopen it and click **fortnight** — three selections, back to back, without pausing to let any of them finish loading.

⚠ The key part is **not waiting**. Each click should happen before the previous click's pricing call has returned (well under a second per click). Firing the same digit into dev tools' Network tab, each call to `ActionCallPricingEngineNZAA` takes roughly 150–250 ms — that's the window to click inside.

Stop clicking and wait ~2 seconds for the network to go quiet.

Observe the "Your premium" panel. Expected: it matches **Evidence B** (\$21.15 /fortnight). Actual (when it reproduces): the selector correctly reads /fortnight , but the price shown is \$550.00 — the premium for **year**, the first and slowest of the three requests to resolve (**Evidence C**).

EVIDENCE

AA Life Insurance

Quote

Application

Summary & payment

Contact us

You can choose up to \$1.5 million of cover. The amount you choose will affect your premium.

\$ 500,000.00

Add additional cover

Serious illness

Add cover

Cancer Care Cover

Add cover

Income Protection

Add cover

Funeral Cover For Someone Else

Add cover

Life Cover

\$45.83 /month

✕

➕ Add Promo Code

➕ Add AA Membership number

Save and email quote

0800 AA LIFE

Request a call back

A • BASELINE

SELECTOR	PREMIUM
/month	\$45.83

AA Life Insurance

Quote

Application

Summary & payment

Contact us

Test, here's your quote

This quote is an indication only. Next, we'll ask you a few additional questions — please note that your answers may alter your premiums. This quote is valid for 30 days.

Life Cover

\$21.15 /fortnight

✕ Remove

About Life Cover

▼

Cover amount

You can choose up to \$1.5 million of cover. The amount you choose will affect your premium.

\$ 500,000.00

Quote Number: Q2001519

Your premium:

\$21.15 /fortnight

AA Member premium:

\$20.09 /fortnight

AA MEMBERSHIP PRICE

Save 5% by providing a valid AA Membership number

join now

B • EXPECTED

SELECTOR	PREMIUM
/fortnight	\$21.15

AA Life Insurance

Quote

Application

Summary & payment

Contact us

Test, here's your quote

This quote is an indication only. Next, we'll ask you a few additional questions — please note that your answers may alter your premiums. This quote is valid for 30 days.

Life Cover

\$550.00 /fortnight

✕ Remove

About Life Cover

▼

Cover amount

You can choose up to \$1.5 million of cover. The amount you choose will affect your premium.

\$ 500,000.00

Quote Number: Q2001519

Your premium:

\$550.00 /fortnight

AA Member premium:

\$522.50 /fortnight

AA MEMBERSHIP PRICE

Save 5% by providing a valid AA Membership number

join now

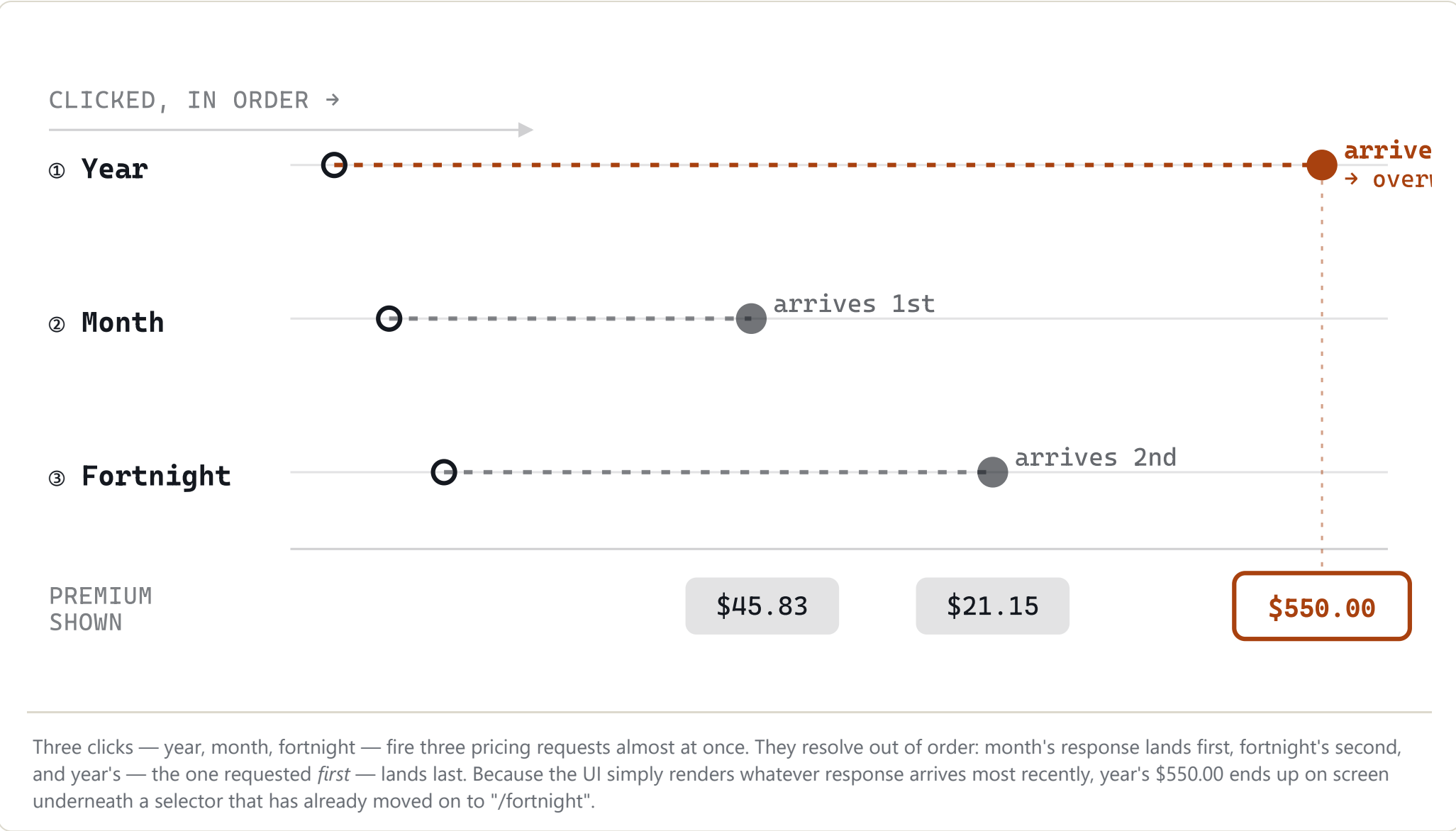
C • ACTUAL (BUG)

SELECTOR	PREMIUM
/fortnight	\$550.00

MISMATCH

ROOT CAUSE

All three clicks in step 3 fire their own call to the pricing engine. Nothing in the UI cancels a request when the selection changes again, and nothing checks — when a response finally arrives — whether it still matches what's currently selected. The last response to *land* simply overwrites the premium, regardless of whether it's the last one the user actually asked for.



REFERENCE VALUES (\$500,000 LIFE COVER)

FREQUENCY	PREMIUM	AA MEMBER PREMIUM
Month	\$45.83	\$43.54
Fortnight	\$21.15	\$20.09
Quarter	\$137.50	—
Half year	\$275.00	—
Year	\$550.00	\$522.50
Shown in Evidence C, mislabelled as "/fortnight"	\$550.00	\$522.50

Note the mismatched state's "AA Member premium" (\$522.50) is exactly 95% of \$550.00, matching the normal 5% AA-member discount applied elsewhere — both figures on screen are internally consistent with **year**, confirming the whole pricing block rendered from a single stale response rather than one widget lagging behind another.

REPRODUCIBILITY

This is timing-dependent, not deterministic on every attempt. In automated testing, rapid 3-to-6-click sequences fired without waiting for the pricing call reproduced a label/premium mismatch in roughly 2 of every 5 attempts, across multiple separate runs. Every reproduction observed so far happened on a sequence that ended with **fortnight** as the final selection — worth a closer look in case fortnight's request/response handling differs from the other frequencies.

No loading indicator is shown while a recalculation is in flight, so a user has no visual cue that the number on screen might still change underneath them.

SUGGESTED FIX

- Guard** Tag each pricing request with the frequency (or a monotonic sequence number) it was made for, and when a response arrives, discard it if it doesn't match the currently-selected frequency.
- Cancel** Use `AbortController` (or the app framework's equivalent) to cancel the in-flight pricing call whenever the user changes the selection again before it resolves.
- Signal** Show a brief loading/disabled state on the premium while a recalculation is pending, so it's visually clear the number isn't final yet — this both helps users and would have made this bug obvious during manual testing.

TEST ARTIFACT

This is automated in `payment-period-staleness.spec.js`. It drives the flow with placeholder data, establishes a correct ground-truth premium for every frequency via the slow path, then fires several rapid frequency-switch sequences and fails if the settled premium doesn't match the frequency label shown.

```
npx playwright test payment-period-staleness.spec.js
```